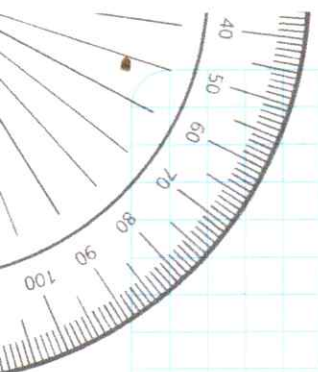
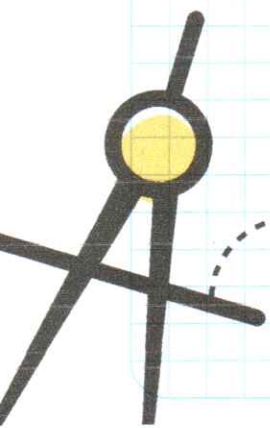
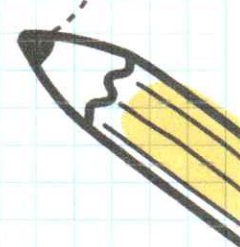




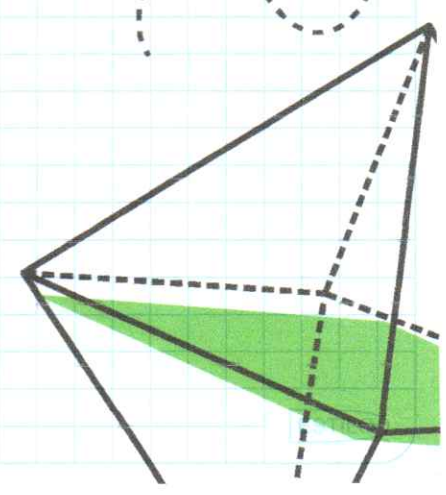
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Trabajos individuales



MATEMÁTICA III





Datos Informativos

Nombre del estudiante: Loor Marvin

Docente: Msc. Diego Tipán

Asignatura: Matemáticas III

Curso: Intermedio

Tema: 25 Ejercicios variados de cálculo diferencial

Fecha: 10-09-26

① $F(x) = 5x^4 - 3x^2 + 7x - 8$

$$F'(x) = \frac{d}{dx}(5x^4) - \frac{d}{dx}(3x^2) + \frac{d}{dx}(7x) - \frac{d}{dx}(8)$$

$$F'(x) = 4(5x^3) - 2(3x) + 1(7x^0) - 0$$

$$F'(x) = 20x^3 - 6x + 7 //$$

③ $F(x) = \sqrt{x} + x^{-2}$; $F(x) = x^{1/2} + x^{-2}$

$$F'(x) = \frac{d}{dx}(x^{1/2}) + \frac{d}{dx}(x^{-2})$$

$$F'(x) = \frac{1}{2}(x^{-1/2}) - 2(x^{-3})$$

$$F'(x) = \frac{1}{2\sqrt{x}} - \frac{2}{x^3} //$$

② $F(x) = (3x^2 + 1)(x^3 - 4)$ | $F'(x) = u^v v + v^u u$

$$F'(x) = \frac{d}{dx}(3x^2 + 1)(x^3 - 4) + \frac{d}{dx}(x^3 - 4)(3x^2 + 1)$$

$$F'(x) = (6x + 0)(x^3 - 4) + (3x^2)(3x^2 + 1)$$

$$F'(x) = 6x^4 - 24x + 9x^4 + 3x^2$$

$$F'(x) = 15x^4 + 3x^2 - 24x //$$

④ $f(x) = \frac{x^2 + 3}{x - 1}$ | $F'(x) = \frac{u^v v - v^u u}{v^2}$

$$F'(x) = \frac{\frac{d}{dx}(x^2 + 3)(x - 1) - \frac{d}{dx}(x - 1)(x^2 + 3)}{(x - 1)^2}$$

$$F'(x) = \frac{(2x)(x - 1) - (1)(x^2 + 3)}{(x - 1)^2}$$

$$F'(x) = \frac{2x^2 - 2x - x^2 - 3}{(x - 1)^2}$$

$$F'(x) = \frac{x^2 - 2x - 3}{(x - 1)^2} //$$

⑤ $F(x) = (2x + 5)^6$

$$F'(x) = \frac{d}{dx}[(2x + 5)^6]$$

$$F'(x) = 6(2x + 5)^5 \cdot \frac{d}{dx}(2x + 5)$$

$$F'(x) = 6(2x + 5)^5 \cdot (2)$$

$$F'(x) = 12(2x + 5)^5 //$$

⑥ $F(x) = \sqrt{3x^2 + 1}$; $F(x) = (3x^2 + 1)^{1/2}$

$$F'(x) = \frac{d}{dx}[(3x^2 + 1)^{1/2}]$$

$$F'(x) = \frac{1}{2}(3x^2 + 1)^{-1/2} \cdot \frac{d}{dx}(3x^2 + 1)$$

$$F'(x) = \frac{1}{2}(3x^2 + 1)^{-1/2} \cdot (6x)$$

$$F'(x) = \frac{1 \cdot 6x}{2(3x^2 + 1)^{1/2}} \cdot \frac{3x}{1}$$

$$F'(x) = \frac{3x}{\sqrt{3x^2 + 1}} //$$



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⑧ $F(x) = \sin x + 3 \cos x$

$$F'(x) = D_x(\sin x) + 3 \cdot D_x(\cos x)$$

$$F'(x) = \cos x + 3(-\sin x)$$

$$F'(x) = \cos x - 3 \sin x //$$

⑦ $F(x) = \sin(4x)$

$$F'(x) = D_x[\sin(4x)]$$

$$F'(x) = \cos(4x) \cdot D_x(4x)$$

$$F'(x) = \cos(4x) \cdot (4)$$

$$F'(x) = 4 \cos(4x) //$$

⑨ $F(x) = \tan(x^2)$

$$F'(x) = D_x[\tan(x^2)]$$

$$F'(x) = \sec^2(x^2) \cdot D_x(x^2)$$

$$F'(x) = \sec^2(x^2) \cdot (2x)$$

$$F'(x) = 2x \sec^2(x^2) //$$

⑩ $F(x) = e^{3x}$

$$F'(x) = D_x[e^{3x}] = e^{3x} \cdot D_x(3x)$$

$$F'(x) = e^{3x} \cdot 3$$

$$F'(x) = 3e^{3x} //$$

⑪ $F(x) = x^2 e^x$

$$F'(x) = D_x(x^2)(e^x) + D_x(e^x)(x^2)$$

$$F'(x) = (2x)(e^x) + e^x x^2$$

$$F'(x) = e^x(2x + x^2) //$$

⑫ $F(x) = \ln(x^2 + 1)$

$$F'(x) = D_x[\ln(x^2 + 1)]$$

$$F'(x) = \frac{1}{(x^2 + 1)} \cdot D_x(x^2 + 1)$$

$$F'(x) = \frac{1}{(x^2 + 1)} \cdot 2x$$

$$F'(x) = \frac{2x}{(x^2 + 1)} //$$

⑬ $F(x) = x \ln x$

$$F'(x) = D_x(x)(\ln x) + D_x(\ln x)(x)$$

$$F'(x) = 1(\ln x) + \left(\frac{1}{x} \cdot D_x(x)\right)(x)$$

$$F'(x) = \ln x + \frac{x}{x}$$

$$F'(x) = \ln x + 1 //$$

⑭ $F(x) = \frac{\ln x}{x}$

$$F'(x) = \frac{D_x(\ln x)(x) - D_x(x)(\ln x)}{x^2}$$

$$F'(x) = \frac{\frac{1}{x}(x) - (1)(\ln x)}{x^2}$$

$$F'(x) = \frac{1 - \ln x}{x^2} //$$

$$(15) f(x) = \sin(3x^2)$$

$$f'(x) = D_x[\sin(3x^2)]$$

$$f'(x) = \cos(3x^2) \cdot D_x(3x^2)$$

$$f'(x) = \cos(3x^2) \cdot 6x$$

$$f'(x) = 6x \cos(3x^2)$$

$$(16) f(x) = \cos(5x-2)$$

$$f'(x) = D_x[\cos(5x-2)]$$

$$f'(x) = -\sin(5x-2) \cdot D_x(5x-2)$$

$$f'(x) = -\sin(5x-2) \cdot (5)$$

$$f'(x) = -5\sin(5x-2)$$

$$(17) f(x) = \tan(x^3+4x)$$

$$f'(x) = D_x[\tan(x^3+4x)]$$

$$f'(x) = \sec^2(x^3+4x) \cdot D_x(x^3+4x)$$

$$f'(x) = \sec^2(x^3+4x) (3x^2+4)$$

$$f'(x) = (3x^2+4) \sec^2(x^3+4x)$$

$$(18) f(x) = \sin(\sqrt{x}) ; f(x) = \sin(x^{1/2})$$

$$f'(x) = D_x[\sin(x^{1/2})]$$

$$f'(x) = \cos(x^{1/2}) \cdot D_x(x^{1/2})$$

$$f'(x) = \cos(x^{1/2}) \left(\frac{1}{2}x^{-1/2}\right)$$

$$f'(x) = \cos(x^{1/2}) \left(\frac{1}{2} \cdot \frac{1}{x^{1/2}}\right)$$

$$f'(x) = \cos(x^{1/2}) \left(\frac{1}{2x^{1/2}}\right)$$

$$f'(x) = \frac{\cos(\sqrt{x})}{2\sqrt{x}}$$

$$(19) f(x) = \cos[(2x+1)^3]$$

$$f'(x) = D_x[\cos[(2x+1)^3]]$$

$$f'(x) = -\sin[(2x+1)^3] \cdot D_x[(2x+1)^3]$$

$$f'(x) = -\sin[(2x+1)^3] \cdot [3(2x+1)^2] \cdot D_x(2x+1)$$

$$f'(x) = -\sin[(2x+1)^3] \cdot [3(2x+1)^2] \cdot (2)$$

$$f'(x) = -6\sin[(2x+1)^3] (2x+1)^2$$

$$f'(x) = -6(2x+1)^2 \sin[(2x+1)^3]$$

$$(20) f(x) = \cos(\sqrt{2x^2+3x}) ; f(x) = \cos[(2x^2+3x)^{1/2}]$$

$$f'(x) = D_x[\cos[(2x^2+3x)^{1/2}]]$$

$$f'(x) = -\sin[(2x^2+3x)^{1/2}] \cdot D_x[(2x^2+3x)^{1/2}]$$

$$f'(x) = -\sin[(2x^2+3x)^{1/2}] \cdot \left[\frac{1}{2}(2x^2+3x)^{-1/2}\right] \cdot D_x(2x^2+3x)$$

$$f'(x) = -\sin[(2x^2+3x)^{1/2}] \cdot \left[\frac{1}{2}(2x^2+3x)^{-1/2}\right] \cdot (4x+3)$$

$$f'(x) = -\sin[(2x^2+3x)^{1/2}] \cdot \left[\frac{1}{2} \cdot \frac{1}{(2x^2+3x)^{1/2}}\right] \cdot (4x+3)$$

$$f'(x) = -\sin[(2x^2+3x)^{1/2}] \cdot \frac{(4x+3)}{2\sqrt{2x^2+3x}}$$

$$f'(x) = \frac{-\sin(\sqrt{2x^2+3x}) (4x+3)}{2\sqrt{2x^2+3x}}$$

$$f'(x) = \frac{-(4x+3) \sin(\sqrt{2x^2+3x})}{2\sqrt{2x^2+3x}}$$

$$(21) \int (6x^5 - 4x^3 + 7x^2 - 9) dx$$

$$= 6 \int x^5 dx - 4 \int x^3 dx + 7 \int x^2 dx - 9 \int dx$$

$$= \frac{6x^{5+1}}{5+1} - \frac{4x^{3+1}}{3+1} + \frac{7x^{2+1}}{2+1} - \frac{9x}{1} + C$$

$$= \frac{6x^6}{6} - \frac{4x^4}{4} + \frac{7x^3}{3} - 9x + C$$

$$= x^6 - x^4 + \frac{7}{3}x^3 - 9x + C$$

$$(22) \int (5x^4 + 3x^2 - \frac{2}{x^3}) dx$$

$$= \int (5x^4 + 3x^2 - 2x^{-3}) dx$$

$$= 5 \int x^4 dx + 3 \int x^2 dx - 2 \int x^{-3} dx$$

$$= \frac{5x^{4+1}}{4+1} + \frac{3x^{2+1}}{2+1} - \frac{2x^{-3+1}}{-3+1} + C$$

$$= \frac{5x^5}{5} + \frac{3x^3}{3} - \frac{2x^{-2}}{-2} + C$$

$$= x^5 + x^3 + x^{-2} + C$$

$$= x^5 + x^3 + \frac{1}{x^2} + C$$

$$(23) \int (3x^2 - 2x + 4)(x) dx$$

$$= \int (3x^3 - 2x^2 + 4x) dx$$

$$= 3 \int x^3 dx - 2 \int x^2 dx + 4 \int x dx$$

$$= \frac{3x^{3+1}}{3+1} - \frac{2x^{2+1}}{2+1} + \frac{4x^{1+1}}{1+1} + C$$

$$= \frac{3x^4}{4} - \frac{2x^3}{3} + \frac{4x^2}{2} + C$$

$$= \frac{3}{4}x^4 - \frac{2}{3}x^3 + 2x^2 + C$$

$$(24) \int (4x^3 + 3\sqrt{x} - \frac{5}{\sqrt{x}}) dx$$

$$= \int (4x^3 + 3x^{1/2} - 5x^{-1/2}) dx$$

$$= 4 \int x^3 dx + 3 \int x^{1/2} dx - 5 \int x^{-1/2} dx$$

$$= \frac{4x^{3+1}}{3+1} + \frac{3x^{1/2+1}}{\frac{1}{2}+1} - \frac{5x^{-1/2+1}}{-\frac{1}{2}+1} + C$$

$$= \frac{4x^4}{4} + \frac{3x^{3/2}}{\frac{3}{2}} - \frac{5x^{1/2}}{\frac{1}{2}} + C$$

$$= x^4 + \frac{6x^{3/2}}{3} - \frac{10x^{1/2}}{1} + C$$

$$= x^4 + 2\sqrt{x^3} - 10\sqrt{x} + C$$

$$(25) \int (\frac{3x^2}{\sqrt{x}} - \frac{4\sqrt{x}}{x^2} + \frac{5}{2\sqrt{x}}) dx$$

$$= \int (3x^{3/2} - 4x^{-3/2} + \frac{5}{2}x^{-1/2}) dx$$

$$= 3 \int x^{3/2} dx - 4 \int x^{-3/2} dx + \frac{5}{2} \int x^{-1/2} dx$$

$$= \frac{3x^{3/2+1}}{\frac{3}{2}+1} - \frac{4x^{-3/2+1}}{-\frac{3}{2}+1} + \frac{5}{2} \cdot \frac{x^{-1/2+1}}{-\frac{1}{2}+1} + C$$

$$= \frac{3x^{5/2}}{\frac{5}{2}} - \frac{4x^{-1/2}}{-\frac{1}{2}} + \frac{5x^{1/2}}{-1}$$

$$= \frac{6x^{5/2}}{5} + 8x^{-1/2} - 5x^{1/2} + C$$

$$= \frac{6\sqrt{x^5}}{5} + \frac{3}{\sqrt{x}} + C$$



Datos Informativos

Nombre del estudiante: Loor Marvin

Docente: Msc. Diego Tipán

Asignatura: Matemática III

Curso: Interciclo

Tema: 20 ejercicios Unidad 1

Fecha: 14-09-26

Tema 1: Antiderivada o primitiva de una función

① $F'(x) = 6x^5 - 8x^3 + 3x^2 - 7$

$$\begin{aligned} F(x) &= \int (6x^5 - 8x^3 + 3x^2 - 7) dx \\ &= 6 \int x^5 dx - 8 \int x^3 dx + 3 \int x^2 dx - 7 \int dx \\ &= \frac{6x^{5+1}}{5+1} - \frac{8x^{3+1}}{3+1} + \frac{3x^{2+1}}{2+1} - 7x + C \\ &= \frac{6x^6}{6} - \frac{8x^4}{4} - \frac{3x^3}{3} - 7x + C \\ &= x^6 - 2x^4 - x^3 - 7x + C \end{aligned}$$

② $F(x) = 5\sqrt{x} - \frac{3}{x^2} + \frac{4}{\sqrt{x}}$

$$\begin{aligned} f(x) &= 5x^{1/2} - 3x^{-2} + 4x^{-1/2} \\ F(x) &= \int (5x^{1/2} - 3x^{-2} + 4x^{-1/2}) dx \\ &= 5 \int x^{1/2} dx - 3 \int x^{-2} dx + 4 \int x^{-1/2} dx \\ &= \frac{5x^{1/2+1}}{\frac{1}{2}+1} - \frac{3x^{-2+1}}{-2+1} + \frac{4x^{-1/2+1}}{-\frac{1}{2}+1} + C \\ &= \frac{5x^{3/2}}{\frac{3}{2}} - \frac{3x^{-1}}{-1} + \frac{4x^{1/2}}{\frac{1}{2}} + C \\ &= \frac{10x^{3/2}}{3} + 3x^{-1} + 8x^{1/2} + C \\ &= \frac{10}{3}\sqrt{x^3} + \frac{3}{x} + 8\sqrt{x} + C \end{aligned}$$

③ Si $F(1) = 7$, hallar $F(x)$ sabiendo que:

$$F'(x) = 4x^3 - 6x + 5$$

$$\begin{aligned} F(x) &= \int (4x^3 - 6x + 5) dx \\ &= 4 \int x^3 dx - 6 \int x dx + 5 \int dx \\ &= \frac{4x^{3+1}}{3+1} - \frac{6x^{1+1}}{1+1} + 5x + C \\ &= \frac{4x^4}{4} - \frac{6x^2}{2} + 5x + C \\ &= x^4 - 3x^2 + 5x + C \end{aligned}$$

$$F(1) = (1)^4 - 3(1)^2 + 5(1) + C = 7$$

$$F(1) = 1 - 3 + 5 + C = 7$$

$$F(1) = 3 + C = 7$$

$$F(1) = C = 7 - 3; C = 4$$

$$F(x) = x^4 - 3x^2 + 5x + 4$$

④ $F(x) = \frac{2x^4 - 3x^2 + 5}{x^2}$

$$\begin{aligned} F(x) &= \frac{2x^4}{x^2} - \frac{3x^2}{x^2} + \frac{5}{x^2} \\ F(x) &= 2x^2 - 3 + \frac{5}{x^2} \end{aligned}$$

$$\begin{aligned} F(x) &= \int (2x^2 - 3 + 5x^{-2}) dx \\ &= 2 \int x^2 dx - 3 \int dx + 5 \int x^{-2} dx \\ &= \frac{2x^{2+1}}{2+1} - 3x + \frac{5x^{-2+1}}{-2+1} + C \\ &= \frac{2x^3}{3} - 3x + \frac{5x^{-1}}{-1} + C \\ &= \frac{2}{3}x^3 - 3x - \frac{5}{x} + C \end{aligned}$$



Datos Informativos

(5) Hallar $F(x)$ si: $F'(x) = 3x^{2/3} - 2x^{-3/2} + 6$

$$F(x) = \int (3x^{2/3} - 2x^{-3/2} + 6) dx$$

$$= 3 \int x^{2/3} dx - 2 \int x^{-3/2} dx + 6 \int dx$$

$$= \frac{3x^{2/3+1}}{\frac{2}{3}+1} - \frac{2x^{-3/2+1}}{-\frac{3}{2}+1} + 6x + C$$

$$= \frac{3x^{5/3}}{\frac{5}{3}} - \frac{2x^{-1/2}}{-\frac{1}{2}} + 6x + C$$

$$= \frac{9x^{5/3}}{5} + 4x^{-1/2} + 6x + C$$

$$= \frac{9}{5} \sqrt[3]{x^5} + \frac{4}{\sqrt{x}} + 6x + C$$

Tema 2. Integrales indefinidas de funciones polinómicas

(6) $\int (7x^6 - 5x^4 + 3x^2 - 8x + 9) dx$

$$= 7 \int x^6 dx - 5 \int x^4 dx + 3 \int x^2 dx - 8 \int x dx + 9 \int dx$$

$$= \frac{7x^{6+1}}{6+1} - \frac{5x^{4+1}}{4+1} + \frac{3x^{2+1}}{2+1} - \frac{8x^{1+1}}{1+1} + 9x + C$$

$$= \frac{7x^7}{7} - \frac{5x^5}{5} + \frac{3x^3}{3} - \frac{8x^2}{2} + 9x + C$$

$$= x^7 - x^5 + x^3 - 4x^2 + 9x + C$$

(7) $\int (3x-2)(x^2+4x-5) dx$

$$= \int (3x^3 + 12x^2 - 15x - 2x^2 - 8x + 10) dx$$

$$= \int (3x^3 + 10x^2 - 23x + 10) dx$$

$$= 3 \int x^3 dx + 10 \int x^2 dx - 23 \int x dx + 10 \int dx$$

$$= \frac{3x^{3+1}}{3+1} + \frac{10x^{2+1}}{2+1} - \frac{23x^{1+1}}{1+1} + 10x + C$$

$$= \frac{3x^4}{4} + \frac{10x^3}{3} - \frac{23x^2}{2} + 10x + C$$

$$= \frac{3}{4}x^4 + \frac{10}{3}x^3 - \frac{23}{2}x^2 + 10x + C$$

(8) $\int \frac{4x^5 - 6x^3 + 2x^2 - 8}{2x^2} dx$; $\frac{4x^5}{2x^2} - \frac{6x^3}{2x^2} + \frac{2x^2}{2x^2} - \frac{8}{2x^2}$; $2x^3 - 3x + 1 - 4x^{-2}$

$$= \int (2x^3 - 3x + 1 - 4x^{-2}) dx$$

$$= 2 \int x^3 dx - 3 \int x dx + 1 \int dx - 4 \int x^{-2} dx$$

$$= \frac{2x^{3+1}}{3+1} - \frac{3x^{1+1}}{1+1} + x - \frac{4x^{-2+1}}{-2+1} + C$$

$$= \frac{2x^4}{4} - \frac{3x^2}{2} + x - \frac{4x^{-1}}{-1} + C$$

$$= \frac{1}{2}x^4 - \frac{3}{2}x^2 + x + 4x^{-1} + C$$

$$= \frac{x^4}{2} - \frac{3}{2}x^2 + x + \frac{4}{x} + C$$

$$\begin{aligned}
 (9) \int (6x^{5/2} - 4x^{3/2} + 3x^{1/2}) dx \\
 &= 6 \int x^{5/2} dx - 4 \int x^{3/2} dx + 3 \int x^{1/2} dx \\
 &= \frac{6x^{5/2+1}}{\frac{5}{2}+1} - \frac{4x^{3/2+1}}{\frac{3}{2}+1} + \frac{3x^{1/2+1}}{-\frac{1}{2}+1} + C \\
 &= \frac{6x^{7/2}}{\frac{7}{2}} - \frac{4x^{5/2}}{\frac{5}{2}} + \frac{3x^{3/2}}{\frac{1}{2}} + C \\
 &= \frac{12x^{7/2}}{7} - \frac{8x^{5/2}}{5} + 6x^{3/2} + C \\
 &= \frac{12}{7}\sqrt{x^7} - \frac{8}{5}\sqrt{x^5} + 6\sqrt{x} + C
 \end{aligned}$$

$$\begin{aligned}
 (10) \int \frac{3x^4\sqrt{x} - 2x^3 + 5\sqrt{x}}{\sqrt{x}} dx \\
 &= \int \frac{3x^4\sqrt{x}}{\sqrt{x}} dx - \int \frac{2x^3}{\sqrt{x}} dx + \int \frac{5\sqrt{x}}{\sqrt{x}} dx \\
 &= \int 3x^4 dx - \int 2x^3 \cdot x^{1/2} dx + \int 5 dx \\
 &= 3 \int x^4 dx - 2 \int x^{5/2} dx + 5 \int dx \\
 &= \frac{3x^{4+1}}{4+1} - \frac{2x^{5/2+1}}{\frac{5}{2}+1} + 5x + C \\
 &= \frac{3x^5}{5} - \frac{2x^{7/2}}{\frac{7}{2}} + 5x + C \\
 &= \frac{3}{5}x^5 - \frac{4x^{7/2}}{7} + 5x + C \\
 &= \frac{3}{5}x^5 - \frac{4}{7}\sqrt{x^7} + 5x + C
 \end{aligned}$$

Tema 3: Integrales indefinidas trascendentales

$$\begin{aligned}
 (11) \int (4e^x - 3\sin x + 5\cos x) dx \\
 &= 4 \int e^x dx - 3 \int \sin x dx + 5 \int \cos x dx \\
 &= 4e^x - 3(-\cos x) + 5\sin x + C \\
 &= 4e^x + 3\cos x + 5\sin x + C
 \end{aligned}$$

$$\begin{aligned}
 (12) \int (5e^{3x} - 4\cos(2x)) dx ; \int e^{ax} dx = \frac{1}{a}e^{ax} \\
 &= 5 \int e^{3x} dx - 4 \int \cos(2x) dx \\
 &= 5 \int e^{3x} dx - 4 \int \cos u \cdot \frac{du}{2} \quad \begin{array}{l} u = 2x \\ du = 2 \cdot dx \\ \frac{du}{2} = dx \end{array} \\
 &= 5 \int e^{3x} dx - \frac{4}{2} \int \cos u du \\
 &= 5 \int e^{3x} dx - 2 \int \cos u du \\
 &= 5 \cdot \frac{1}{3} e^{3x} - 2 \sin u + C \\
 &= \frac{5}{3} e^{3x} - 2 \sin(2x) + C
 \end{aligned}$$

$$\begin{aligned}
 (13) \int (3\sec^2(4x) + 5\csc^2(2x)) dx \\
 &= 3 \int \sec^2(4x) dx + 5 \int \csc^2(2x) dx \\
 &= 3 \int \sec^2(u) \cdot \frac{du}{4} + 5 \int \csc^2(v) \cdot \frac{dv}{2} \\
 &= \frac{3}{4} \int \sec^2 u du + \frac{5}{2} \int \csc^2 v dv \\
 &= \frac{3}{4} \tan u - \frac{5}{2} \cot v + C \\
 &= \frac{3}{4} \tan(4x) - \frac{5}{2} \cot(2x) + C
 \end{aligned}$$

$$(14) \int \left(\frac{4}{x} + 3e^{-2x} - 2\sin(5x) \right) dx$$

$$\int (4x^{-1} + 3e^{-2x} - 2\sin(5x)) dx$$

$$u = 5x \\ du = 5 \cdot dx \\ \frac{du}{5} = dx$$

$$4 \int x^{-1} dx + 3 \int e^{-2x} dx - 2 \int \sin u \cdot \frac{du}{5}$$

$$4 \int x^{-1} dx + 3 \int e^{-2x} dx - \frac{2}{5} \int \sin u du$$

$$= 4 \ln|x| + 3 \cdot \frac{1}{-2} e^{-2x} + \frac{2}{5} \cos u + C$$

$$= 4 \ln|x| - \frac{3}{2} e^{-2x} + \frac{2}{5} \cos u + C$$

$$= 4 \ln|x| - \frac{3}{2} e^{-2x} + \frac{2}{5} \cos(5x) + C$$

$$(15) \int (2^x + 4\sec(3x) + \tan(3x) - 6\csc(2x)\cot(2x)) dx$$

$$\int 2^x dx + 4 \int \sec u \tan u \cdot \frac{du}{3} - 6 \int \csc v \cot v \cdot \frac{dv}{2}$$

$$\int 2^x dx + \frac{4}{3} \int (\sec u \tan u) du - \frac{6}{2} \int (\csc v \cot v) dv$$

$$\int 2^x dx + \frac{4}{3} \int (\sec u \tan u) du - 3 \int (\csc v \cot v) dv$$

$$= \frac{2^x}{\ln 2} + \frac{4}{3} \sec u + 3 \csc v + C$$

$$= \frac{2^x}{\ln 2} + \frac{4}{3} \sec(3x) + 3 \csc(2x) + C$$

Tema 4: Integrales indefinidas inversas e hiperbólicas

$$(16) \int \frac{1}{x^2 + 25} dx ; \int \frac{1}{x^2 + 5^2} dx$$

$$= \frac{1}{5} \arctan\left(\frac{x}{5}\right) + C$$

$$= \frac{1}{5} \arctan\left(\frac{x}{5}\right) + C$$

$$(17) \int \frac{1}{\sqrt{16-x^2}} dx ; \int \frac{1}{\sqrt{4^2-x^2}} dx$$

$$= \arcsen\left(\frac{x}{4}\right) + C$$

$$= \arcsen\left(\frac{x}{4}\right) + C$$

$$(18) \int \frac{3}{9 + 4x^2} dx$$

$$= \int \frac{3}{3^2 + (2x)^2} dx$$

$$u = 2x \\ du = 2 \cdot dx \\ \frac{du}{2} = dx$$

$$= \int \frac{3}{3^2 + u^2} \cdot \frac{du}{2}$$

$$= \frac{3}{2} \int \frac{1}{3^2 + u^2} du$$

$$= \frac{3}{2} \left[\frac{1}{3} \arctan\left(\frac{u}{3}\right) \right] + C$$

$$= \frac{1}{2} \arctan\left(\frac{2x}{3}\right) + C$$

$$(19) \int (4\sinh(2x) - 3\cosh(5x)) dx$$

$$= 4 \int \sinh(2x) dx - 3 \int \cosh(5x) dx$$

$$= 4 \int \sinh(u) \cdot \frac{du}{2} - 3 \int \cosh(v) \cdot \frac{dv}{5}$$

$$= 2 \int \sinh(u) du - \frac{3}{5} \int \cosh(v) dv$$

$$= 2 \cosh u - \frac{3}{5} \sinh v + C$$

$$= 2 \cosh(2x) - \frac{3}{5} \sinh(5x) + C$$

$$(20) \int \left[\frac{4}{x^2+16} + 3 \cosh(2x) - \frac{2}{\sqrt{9-x^2}} \right] dx$$

$$u = 2x \\ du = 2 \cdot \frac{dx}{2}$$

$$= \int \frac{4}{x^2+4^2} dx + 3 \int \cosh(2x) dx - 2 \int \frac{1}{\sqrt{3^2-x^2}} dx$$

$$= \int \frac{4}{x^2+4^2} dx + 3 \int \cosh u \cdot \frac{du}{2} - 2 \int \frac{1}{\sqrt{3^2-x^2}} dx$$

$$= 4 \int \frac{1}{x^2+4^2} dx + \frac{3}{2} \int \cosh u du - 2 \int \frac{1}{\sqrt{3^2-x^2}} dx$$

$$= 4 \cdot \frac{1}{4} \arctan\left(\frac{x}{4}\right) + \frac{3}{2} \sinh u - 2 \arcsen\left(\frac{x}{3}\right) + C$$

$$= \arctan\left(\frac{x}{4}\right) + \frac{3}{2} \sinh(2x) - 2 \arcsen\left(\frac{x}{3}\right) + C$$



Datos Informativos

Nombre del estudiante: Loor Iváñez Marín Aldahir

Docente: Msc. Diego Tipán

Asignatura: Matemática III

Curso: Intermedio

Tema: Ejercicios Unidad 2

Fecha: 21-09-26

7 ejercicios por método de sustitución

① $\int x(x^2+5)^4 dx$

$$u = x^2 + 5$$
$$\frac{du}{dx} = 2x$$
$$\frac{du}{2} = x dx$$

$$\int (x^2+5)^4 x dx$$

$$\frac{1}{2} \int (u)^4 du$$

$$\frac{1}{2} \int u^4 du$$

$$\frac{1}{2} \cdot \frac{u^{4+1}}{4+1} + C$$

$$= \frac{u^5}{10} + C$$

$$= \frac{(x^2+5)^5}{10} + C$$

② $\int \frac{3x^2}{x^3+7} dx$

$$u = x^3 + 7$$
$$du = 3x^2 dx$$

$$\int \frac{du}{u}$$

$$= \ln|u| + C$$

$$= \ln|x^3+7| + C$$

③ $\int 2x e^{x^2+1} dx$

$$u = x^2 + 1$$
$$du = 2x dx$$

$$\int e^{x^2+1} 2x dx$$

$$\int e^u du$$

$$= e^u + C$$

$$= e^{x^2+1} + C$$

④ $\int \frac{x}{\sqrt{x^2+9}} dx$

$$u = x^2 + 9$$
$$\frac{du}{dx} = 2x$$
$$\frac{du}{2} = x dx$$

$$\int \frac{1}{\sqrt{u}} \cdot \frac{du}{2}$$

$$\frac{1}{2} \int u^{-1/2} du$$

$$= \frac{1}{2} \cdot \frac{u^{-1/2+1}}{-1/2+1} + C$$

$$= \frac{1}{2} \cdot \frac{u^{1/2}}{1/2} + C$$

$$= \frac{1}{2} 2u^{1/2} + C$$

$$= (x^2+9)^{1/2} + C$$

⑤ $\int \frac{4x^3}{(x^4+2)^2} dx$

$$u = x^4 + 2$$
$$du = 4x^3 dx$$

$$\int \frac{du}{(u)^2}$$

$$= -\frac{1}{x^4+2} + C$$

$$\int u^{-2} du$$

$$= \frac{u^{-2+1}}{-2+1} + C$$

$$= \frac{u^{-1}}{-1} + C$$

$$= -\frac{1}{u} + C$$

⑥ $\int x \sqrt{x^2-3} dx$

$$u = x^2 - 3$$
$$du = 2x dx$$

$$\int (x^2-3)^{1/2} x dx$$

$$\int u^{1/2} \cdot \frac{du}{2}$$

$$\frac{1}{2} \int u^{1/2} du$$

$$\frac{1}{2} \cdot \frac{u^{1/2+1}}{1/2+1} + C$$

$$\frac{1}{2} \cdot \frac{u^{3/2}}{3/2} + C$$

$$\frac{1}{2} \cdot \frac{2u^{3/2}}{3} + C$$

$$\frac{1}{3} u^{3/2} + C$$

$$= \frac{1}{3} (x^2-3)^{3/2} + C$$



Datos Informativos

$$\textcircled{1} \int \frac{x \ln(x^2+4)}{x^2+4} dx$$

$$u = x^2+4$$

$$du = 2x dx$$

$$\int \frac{\ln(x^2+4)}{x^2+4} x dx$$

$$v = \ln u$$

$$dv = \frac{1}{u} du$$

$$\int \frac{\ln u}{u} \cdot \frac{du}{2}$$

$$\frac{1}{2} \int \frac{\ln u}{u} du$$

$$\frac{1}{2} \int v dv$$

$$= \frac{1}{2} \cdot \frac{v^2}{2} + C$$

$$= \frac{1}{4} v^2 + C$$

$$= \frac{1}{4} [\ln(x^2+4)]^2 + C$$

③ Ejercicios por integración por partes $\int u dv = uv - \int v du$

$$\textcircled{1} \int x e^x dx$$

$$u = x$$

$$du = dx$$

$$\int dv = \int e^x dx$$

$$dv = e^x$$

$$\int x e^x dx = x e^x - \int e^x dx$$

$$= x e^x - e^x + C$$

$$= e^x (x-1) + C$$

$$\textcircled{2} \int x \cos x dx$$

$$u = x$$

$$du = dx$$

$$\int dv = \int \cos x dx$$

$$v = \sin x$$

$$\int x \cos x dx = x \sin x - \int \sin x dx$$

$$= x \sin x + \cos x + C$$

$$\textcircled{3} \int x \sin x dx$$

$$u = x$$

$$du = dx$$

$$\int dv = \int \sin x dx$$

$$v = -\cos x$$

$$\int x \sin x dx = -x \cos x + \int \cos x dx$$

$$= -x \cos x + \sin x + C$$

$$\textcircled{4} \int \ln x dx$$

$$\int \ln x \cdot 1 dx$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$\int dv = \int dx$$

$$v = x$$

$$\int \ln x \cdot 1 dx = x \ln x - \int x dx$$

$$= x \ln x - \frac{x^2}{2} + C$$

$$= x \ln x - \frac{x^2}{2} + C$$

$$⑤ \int x \ln x \, dx$$

$$u = \ln x \\ du = \frac{1}{x} dx$$

$$\int dv = \int x \, dx \\ v = \frac{x^2}{2}$$

$$\int x \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx$$

$$= \frac{x^2}{2} \ln x - \frac{1}{2} \cdot \frac{x^2}{2} + C$$

$$= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$

$$⑥ \int x^2 e^x \, dx$$

$$u = x^2 \\ du = 2x \, dx$$

$$\int dv = \int e^x \, dx \\ v = e^x$$

$$\int x^2 e^x \, dx = x^2 e^x - \int e^x 2x \, dx$$

$$= e^x (x^2 - 2x + 2) + C$$

$$⑦ \int x^2 \cos x \, dx$$

$$u = x^2 \\ du = 2x \, dx$$

$$\int dv = \int \cos x \, dx \\ v = \sin x$$

$$\int x^2 \cos x \, dx = x^2 \sin x - 2 \int \sin x \, dx$$

$$= x^2 \sin x - 2(-x \cos x + \sin x) + C$$

$$= x^2 \sin x + 2x \cos x - 2 \sin x + C$$

$$⑧ \int x e^{2x} \, dx$$

$$u = x \\ du = dx$$

$$dv = e^{2x} \, dx \\ v = \frac{1}{2} e^{2x}$$

$$\int x e^{2x} \, dx = \frac{x}{2} e^{2x} - \frac{1}{2} \int e^{2x} \, dx$$

$$= \frac{x}{2} e^{2x} - \frac{1}{4} e^{2x} + C$$

$$= \frac{e^{2x}}{4} (2x - 1) + C$$

8 Ejercicios de funciones racionales

$$① \int \frac{5x+1}{x^2+3x+2} \, dx = \int \frac{5x+1}{(x+1)(x+2)} \, dx \quad ; \quad \frac{5x+1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

$$\int \frac{-4}{x+1} \, dx + \int \frac{9}{x+2} \, dx$$

$$-4 \int \frac{dx}{x+1} + 9 \int \frac{dx}{x+2}$$

$$= -4 \ln|x+1| + 9 \ln|x+2| + C$$

$$5x+1 = \frac{A(x+2)}{x+2} + \frac{B(x+1)}{x+1}$$

$$5(-2)+1 = B(-1)$$

$$-9 = B(-1)$$

$$\boxed{a = 9}$$

$$-5+1 = A(1)$$

$$\boxed{-4 = A}$$

$$\textcircled{2} \int \frac{x+4}{x^2-x-6} dx ; \int \frac{x+4}{(x-3)(x+2)} dx \quad \left| \frac{x+4}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2} \right.$$

$$x+4 = A(x+2) + B(x-3)$$

$$-2+4 = B(-5)$$

$$-2 = B(-5)$$

$$\boxed{\frac{-2}{-5} = B}$$

$$4 = A(5)$$

$$\boxed{\frac{4}{5} = A}$$

$$\frac{7}{5} \int \frac{1}{x-3} dx - \frac{2}{5} \int \frac{1}{x+2} dx$$

$$= \frac{7}{5} \ln|x-3| - \frac{2}{5} \ln|x+2| + C$$

$$\textcircled{3} \int \frac{3x+7}{x^2+5x+6} dx \quad \left| \frac{3x+7}{(x+2)(x+3)} = \frac{A}{x+2} + \frac{B}{x+3} \right.$$

$$3x+7 = A(x+3) + B(x+2)$$

$$-2 = B(-1)$$

$$\boxed{2 = B}$$

$$\frac{1 = A(1)}{1 = A}$$

$$\int \frac{1}{x+2} dx + 2 \int \frac{1}{x+3} dx$$

$$= \ln|x+2| + 2 \ln|x+3| + C$$

$$\textcircled{4} \int \frac{2x+5}{(x+1)(x-2)} dx \quad \left| \frac{2x+5}{(x+1)(x-2)} = \frac{A}{x+1} + \frac{B}{x-2} \right.$$

$$2x+5 = A(x-2) + B(x+1)$$

$$19 = B(3)$$

$$\frac{19}{3} = B$$

$$\boxed{\frac{19}{3} = B}$$

$$3 = A(-3)$$

$$\frac{-3}{3} = A$$

$$\boxed{A = -1}$$

$$-\int \frac{1}{x+1} dx + 3 \int \frac{1}{x-2} dx$$

$$= -\ln|x+1| + 3 \ln|x-2| + C$$

$$\textcircled{5} \int \frac{1}{x^2-4} dx \quad \left| \frac{1}{(x-2)(x+2)} = \frac{A}{x-2} + \frac{B}{x+2} \right.$$

$$1 = A(x+2) + B(x-2)$$

$$1 = B(-4)$$

$$\boxed{-\frac{1}{4} = B}$$

$$1 = A(4)$$

$$\boxed{\frac{1}{4} = A}$$

$$\frac{1}{4} \int \frac{1}{x-2} dx - \frac{1}{4} \int \frac{1}{x+2} dx$$

$$= \frac{1}{4} \ln|x-2| - \frac{1}{4} \ln|x+2| + C$$

$$\textcircled{6} \int \frac{2x+1}{(x+1)^2} dx \quad \left| \frac{2x+1}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2} \right.$$

$$\int \left(\frac{2}{x+1} - \frac{1}{(x+1)^2} \right) dx \quad 2x+1 = A(x+1) + B$$

$$= 2 \ln|x+1| + \frac{1}{x+1} + C \quad \begin{matrix} -2+1 = B \\ -1 = B \end{matrix} \quad \boxed{A=2}$$

$$\textcircled{7} \int \frac{x^2+3}{x(x-1)(x+1)} dx \quad \left| \frac{x^2+3}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1} \right.$$

$$3 \int \frac{dx}{x} + 2 \int \frac{dx}{x-1} + 2 \int \frac{dx}{x+1} \quad x^2+3 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$$

$$= -3 \ln|x| + 2 \ln|x-1| + 2 \ln|x+1| + C \quad \boxed{A=4} \quad \boxed{2=B} \quad \boxed{2=C}$$

$$\textcircled{8} \int \frac{2x^2+3x+4}{x(x+1)(x+2)} dx \quad \left| \frac{2x^2+3x+4}{x(x+1)(x+2)} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+2} \right.$$

$$\int \left(\frac{2}{x} - \frac{3}{x+1} + \frac{3}{x+2} \right) dx \quad 2x^2+3x+4 = A(x+1)(x+2) + Bx(x+2) + Cx(x+1)$$

$$= 2 \ln|x| - 3 \ln|x+1| + 3 \ln|x+2| + C \quad \boxed{4=2A} \quad \boxed{3=-B} \quad \boxed{6=2C}$$

$$\boxed{2=4}$$

$$\boxed{-3=B}$$

$$\boxed{3=C}$$



Datos Informativos

Nombre del estudiante: Loor Tuzrez Marvin Aldahir 120741566-0

Docente: Msc. Diego Tipán

Asignatura: Matemática III

Curso: Intermedio

Tema: Ejercicios Unidad 2, tema 2.

Fecha: 16-09-26

① $\int \frac{1}{\sqrt{x^2+1}} dx$

$x = \tan \theta$
 $dx = \sec^2 \theta \cdot d\theta$

$= \int \frac{1}{\sqrt{(\tan \theta)^2+1}} \cdot \sec^2 \theta d\theta$

$= \int \frac{\sec^2 \theta}{\sqrt{(\tan \theta)^2+1}} d\theta$

$= \int \frac{\sec^2 \theta}{\sec \theta} d\theta$

$= \int \frac{\sec^2 \theta}{\sec \theta} d\theta$

$= \int \sec \theta d\theta$

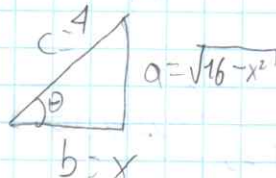
② $\int \frac{dx}{\sqrt{16-x^2}}$

$= \int \frac{4 \cos \theta}{4 \cos \theta} d\theta$

$= \int \theta$

$= \theta + C$

$x = 4 \sin \theta$
 $dx = 4 \cos \theta$



$\cos \theta = \frac{\sqrt{16-x^2}}{4}$

$4 \cos \theta = \sqrt{16-x^2}$

$\sin \theta = \frac{x}{4}$

$4 \sin \theta = x$

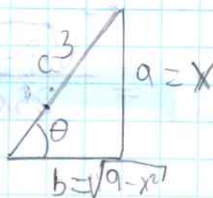
③ $\int \frac{3}{\sqrt{9-x^2}} dx$

$= \int \frac{3}{\sqrt{9-(3 \cos \theta)^2}} \cdot (-3 \sin \theta) d\theta$

$= \int \frac{3}{\sqrt{9-(9 \cos^2 \theta)}} \cdot (-3 \sin \theta) d\theta$

$= \int \frac{3}{\sqrt{9(1-\cos^2 \theta)}} \cdot (-3 \sin \theta) d\theta$

$= \int \frac{3}{\sqrt{9(\sin^2 \theta)}} \cdot (-3 \sin \theta) d\theta$



$\cos \theta = \frac{\sqrt{9-x^2}}{3}$

$3 \cos \theta = x$
 $dx = -3 \sin \theta d\theta$

$a = \frac{b}{c}$

$= \int \frac{3}{3 \sin \theta} \cdot (-3 \sin \theta) d\theta$

$= -3 \int d\theta$

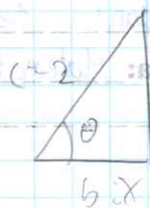
$= -3 \theta + C$

$= -3 \arccos\left(\frac{x}{3}\right) + C$



Datos Informativos

④ $\int \frac{x^2}{\sqrt{4-x^2}} dx$



$\sin \theta = \frac{x}{2}$
 $2 \sin \theta = x$
 $dx = 2 \cos \theta d\theta$

$$\int \frac{(2 \sin \theta)^2}{\sqrt{4 - (2 \sin \theta)^2}} \cdot 2 \cos \theta d\theta$$

$$\int \frac{4 \sin^2 \theta}{\sqrt{4 - (4 \sin^2 \theta)}} \cdot 2 \cos \theta d\theta$$

$$\int \frac{4 \sin^2 \theta}{\sqrt{4(1 - \sin^2 \theta)}} \cdot 2 \cos \theta d\theta$$

$$\int \frac{4 \sin^2 \theta}{\sqrt{4 \cos^2 \theta}} \cdot 2 \cos \theta d\theta$$

$$\int \frac{4 \sin^2 \theta}{2 \cos \theta} \cdot 2 \cos \theta d\theta$$

$$\int (4 \sin^2 \theta) d\theta$$

$$= 4 \int \sin^2 \theta d\theta$$

$$= 4 \int \frac{1 - \cos(2\theta)}{2} d\theta$$

$$= 2 \int (1 - \cos(2\theta)) d\theta$$

$$= 2 \int 1 d\theta - \int \cos(2\theta) d\theta$$

$$= 2 \int d\theta - \int \cos(u) \frac{du}{2}$$

$$= 2 \int d\theta - \frac{1}{2} \int \cos(u) du$$

$$= 2\theta - \frac{1}{2} \sin(2\theta) + C$$

$$= 2 \arcsin\left(\frac{x}{2}\right) - \frac{x \sqrt{4-x^2}}{2} + C$$

$u = 2\theta$
 $\frac{du}{2} = \frac{2 d\theta}{2}$

⑤ $\int \frac{1}{x^2 \sqrt{x^2-25}} dx$

$5 \sec \theta = x$
 $dx = 5 \sec \theta \tan \theta d\theta$

$$\int \frac{1}{(5 \sec \theta)^2 \sqrt{(5 \sec \theta)^2 - 25}} \cdot (5 \sec \theta \tan \theta) d\theta$$

$$\int \frac{1}{25 \sec^2 \theta \sqrt{25 \sec^2 \theta - 25}} \cdot (5 \sec \theta \tan \theta) d\theta$$

$$\int \frac{1}{25 \sec^2 \theta \sqrt{25(\sec^2 \theta - 1)}} \cdot (5 \sec \theta \tan \theta) d\theta$$

$$\int \frac{1}{25 \sec^2 \theta \sqrt{25 \tan^2 \theta}} \cdot (5 \sec \theta \tan \theta) d\theta$$

$$\int \frac{1}{25 \sec^2 \theta (5 \tan \theta)} \cdot (5 \sec \theta \tan \theta) d\theta$$

$$\int \frac{5}{125 \sec^2 \theta \tan \theta} (\sec \theta \tan \theta) d\theta$$

$$\frac{5}{125} \int \frac{1}{\sec \theta} d\theta$$

$$\frac{1}{25} \int \cos \theta d\theta$$

$$= \frac{1}{25} \sin \theta + C$$

$$= \frac{\sqrt{x^2-25}}{25x} + C$$

$$⑥ \int \sqrt{3+x^2} dx$$

$$h = \frac{co}{ca}$$

$$\tan \theta = \frac{3}{x}$$

$$\sqrt{3} \tan \theta = x$$

$$dx = \sqrt{3} \sec^2 \theta d\theta$$

$$\int \sqrt{3 + (\sqrt{3} \tan \theta)^2} \cdot (\sqrt{3} \sec^2 \theta) d\theta$$

$$\int \sqrt{3 + 3 \tan^2 \theta} \cdot (\sqrt{3} \sec^2 \theta) d\theta$$

$$\int \sqrt{3(1 + \tan^2 \theta)} \cdot (\sqrt{3} \sec^2 \theta) d\theta$$

$$\int \sqrt{3 \sec^2 \theta} \cdot (\sqrt{3} \sec^2 \theta) d\theta$$

$$\int (\sqrt{3} \sec \theta \cdot \sqrt{3} \sec^2 \theta) d\theta$$

$$\int 3 \sec^3 \theta d\theta$$

$$\rightarrow 3 \int \sec^3 \theta d\theta$$

$$= \frac{3}{2} \sec \theta \tan \theta + \frac{3}{2} \ln |\sec \theta + \tan \theta| + C$$

$$= \frac{3}{2} \left(\frac{\sqrt{3+x^2}}{\sqrt{3}} \right) \left(\frac{x}{\sqrt{3}} \right) + \frac{3}{2} \ln \left| \left(\frac{\sqrt{3+x^2}}{\sqrt{3}} \right) + \left(\frac{x}{\sqrt{3}} \right) \right| + C$$

$$= \frac{x\sqrt{3+x^2}}{2} + \frac{3}{2} \ln |x + \sqrt{3+x^2}| + C$$

$$⑦ \int \frac{1}{x^2 \sqrt{9-x^2}} dx$$

$$\sin \theta = \frac{x}{3}$$

$$3 \sin \theta = x$$

$$\int \frac{1}{(3 \sin \theta)^2 \sqrt{9 - (3 \sin \theta)^2}} \cdot 3 \cos \theta d\theta = 3 \cos \theta d\theta$$

$$\int \frac{1}{9 \sin^2 \theta \sqrt{9 - 9 \sin^2 \theta}} \cdot 3 \cos \theta d\theta$$

$$\int \frac{1}{9 \sin^2 \theta \sqrt{9(1 - \sin^2 \theta)}} \cdot 3 \cos \theta d\theta$$

$$\int \frac{1}{9 \sin^2 \theta \sqrt{9 \cos^2 \theta}} \cdot 3 \cos \theta d\theta$$

$$\int \frac{1}{9 \sin^2 \theta (3 \cos \theta)} \cdot 3 \cos \theta d\theta$$

$$\int \frac{3 \cos \theta}{27 \sin^2 \theta \cos \theta} d\theta$$

$$\frac{1}{9} \int \frac{1}{\sin^2 \theta} d\theta$$

$$\frac{1}{9} \int \csc^2 \theta d\theta$$

$$= -\frac{1}{9} \cot \theta + C$$

$$= -\frac{\sqrt{9-x^2}}{9x} + C$$



Datos Informativos

Nombre del estudiante: Loor Tudiez Marvin Aldahir
Docente: Msc. Diego Tipán Asignatura: Matemática III Curso: Intermedio
Tema: Ejercicios Variables Unidad 3 Fecha: 28-04-26

Integral definida

① $\int_0^2 (2x+1) dx$

$$\int_0^2 2x + \int_0^2 dx$$
$$\left. \frac{2x^2}{2} + x \right|_0^2$$

$$[(2)^2 - (0)^2] + [2 - 0]$$

$$[4] + [2]$$

$$= 6$$

③ Si $\int_2^7 f(x) dx = 9$, calcula $\int_7^2 f(x) dx$

$$= \int_7^2 f(x) dx$$

$$= - \int_2^7 f(x) dx$$

$$= -9$$

$$= -9$$

⑤ $\int_0^2 (3x^2 + 2x) dx$

$$3 \int_0^2 x^2 dx + 2 \int_0^2 x dx$$

$$\left. x^3 \right|_0^2 + \left. x^2 \right|_0^2$$

$$[(2)^3 - (0)^3] + [(2)^2 - (0)^2]$$

$$= 8 + 4$$

$$= 12$$

② $\int_0^1 x^2 dx$

$$\left. \frac{x^3}{3} \right|_0^1$$
$$= \left[\frac{(1)^3}{3} - \frac{(0)^3}{3} \right]$$

$$= \frac{1}{3} - 0$$

$$= \frac{1}{3}$$

④ $\int_4^9 (x^2 + 3x - 1) dx$

$$= \int_0^9 f(x) dx - \int_0^4 f(x) dx = 0$$

$$= 0$$

⑥ $\int_1^3 (2x^3 - 4x + 5) dx$

$$2 \int_1^3 x^3 dx - 4 \int_1^3 x dx + 5 \int_1^3 dx$$

$$\left. \frac{2x^4}{4} \right|_1^3 - \left. \frac{4x^2}{2} \right|_1^3 + \left. 5x \right|_1^3$$
$$\left. \frac{1}{2} x^4 \right|_1^3 - \left. 2x^2 \right|_1^3 + \left. 5x \right|_1^3$$

$$\frac{1}{2} [(3)^4 - (1)^4] - 2 [(3)^2 - (1)^2] + 5 [3 - 1]$$

$$\frac{1}{2} [81 - 1] - 2 [9 - 1] + 10$$



Datos Informativos

$$\frac{1}{2} [80] - 2[8] + 20$$

$$\frac{80}{2} - 16 + 20$$

$$= 40 - 16$$

$$= 24$$

$$\textcircled{1} \int_1^4 \sqrt{x} dx$$

$$\int_1^4 x^{1/2} dx$$

$$\left[\frac{x^{3/2}}{3/2} \right]_1^4$$

$$= \frac{2}{3} x^{3/2} \Big|_1^4$$

$$= \frac{2}{3} [(4)^{3/2} - (1)^{3/2}]$$

$$\frac{2}{3} [\sqrt{4^3} - \sqrt{1}]$$

$$= \frac{2}{3} [8 - 1]$$

$$= \frac{2}{3} [7]$$

$$= \frac{14}{3}$$

$$= \frac{14}{3}$$

Teorema del valor medio

$$\textcircled{1} \int_0^2 x^2 dx$$

$$f(c) = \frac{1}{2} \int_0^2 x^2 dx$$

$$= \frac{1}{2} \cdot \left[\frac{x^3}{3} \right]_0^2$$

$$= \frac{1}{2} \cdot \frac{1}{3} [2^3 - 0^3]$$

$$= \frac{1}{2} \cdot \frac{1}{3} [8]$$

$$= \frac{1}{2} \cdot \frac{8}{3} = \frac{4}{3}$$

$$= \frac{4}{3}$$

$$f(c) = \frac{4}{3}$$

$$\sqrt{c^2} = \frac{\sqrt{4}}{\sqrt{3}}$$

$$c = \pm \frac{2}{\sqrt{3}}$$

$$c \approx 1.155$$

$$\textcircled{2} \int_1^4 (2x+1) dx$$

$$f(c) = \frac{1}{4-1} \int_1^4 (2x+1) dx$$

$$= \frac{1}{3} \int_1^4 2x dx + \frac{1}{3} \int_1^4 1 dx$$

$$= \frac{1}{3} [x^2 + x]_1^4$$

$$= \frac{1}{3} [(16+4) - (1+1)]$$

$$= \frac{1}{3} (20-2)$$

$$= \frac{18}{3}$$

$$= 6$$

$$c \approx 6$$

$$6 = 2x+1$$

$$5 = 2x$$

$$\frac{5}{2} = x$$

$$x = \frac{5}{2}$$

$$\textcircled{3} \int_1^4 \sqrt{x} dx$$

$$\frac{1}{4-1} \int_1^4 x^{1/2} dx$$

$$\frac{1}{3} \left[\frac{2}{3} x^{3/2} \right]_1^4$$

$$= \frac{1}{3} \left[\frac{2}{3} (8) - \frac{2}{3} (1) \right]$$

$$= \frac{1}{3} \left(\frac{14}{3} \right)$$

$$f(c) = \frac{14}{9}$$

$$\sqrt{c} = \frac{14}{9}$$

$$\sqrt{c^2} = \left(\frac{14}{9} \right)^2$$

$$c = \frac{196}{81}$$

$$c \approx 2.42$$

$$\textcircled{4} F(x) = \int_1^x (t^3 + 2t) dt$$

$$F'(x) = x^3 + 2x$$

$$F'(x) = x^3 + 2x$$

$$⑤ F(x) \int_0^{x^2} (1+t^2) dt$$

$$F'(x) = [1+(x^2)^2] \frac{d}{dx} (x^2)$$

$$= [1+x^4] \cdot (2x)$$

$$= 2x + 2x^5$$

$$⑥ F(x) \int_x^3 \sqrt{1+t^2} dt$$

$$F'(x) = -\sqrt{1+x^2}$$

$$F'(x) = -\sqrt{1+x^2}$$

$$④ \int_1^3 (2x+4) dx$$

$$\int_1^3 2x dx + \int_1^3 4 dx$$

$$[x^2 + 4x]_1^3$$

$$= [(3^2 + 4(3)) - (1^2 + 4(1))]$$

$$[9 + 12 - (1 + 4)]$$

$$[21 - 5]$$

$$= 16 u^2$$

Área entre curvas

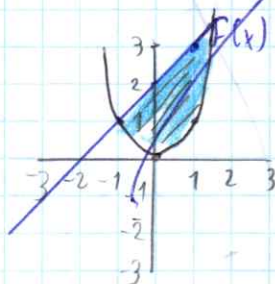
$$① f(x) = x+2 ; g(x) = x^2 ; [0,2]$$

$$f(x)$$

x	y
-1	1
0	2
1	3

$$g(x)$$

x	y
-1	1
0	0
1	1



$$\int_0^2 [f(x) - g(x)] dx$$

$$A = \int_0^2 [(x+2) - x^2] dx$$

$$= \int_0^2 (-x^2 + x + 2) dx$$

$$= \left[-\frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_0^2$$

$$= -\frac{8}{3} + 2 + 4$$

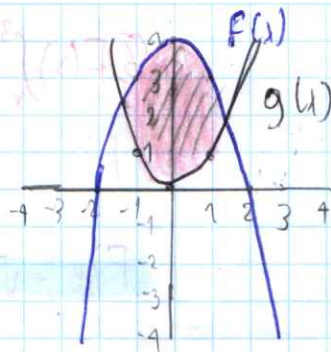
$$= 6 - \frac{8}{3}$$

$$= \frac{10}{3} u^2$$

② $f(x) = 4 - x^2$; $g(x) = x^2$; $[-\sqrt{2}, \sqrt{2}]$

x	y
-1	3
0	4
1	3
2	0

x	y
-1	1
0	0
1	1
2	4



$$\int_{-\sqrt{2}}^{\sqrt{2}} [f(x) - g(x)] dx \rightarrow \frac{1}{2} \left(4\sqrt{2} - \frac{4\sqrt{2}}{3} \right)$$

$$\int_{-\sqrt{2}}^{\sqrt{2}} [(4 - x^2) - (x^2)] dx = 2 \left(\frac{8\sqrt{2}}{3} \right)$$

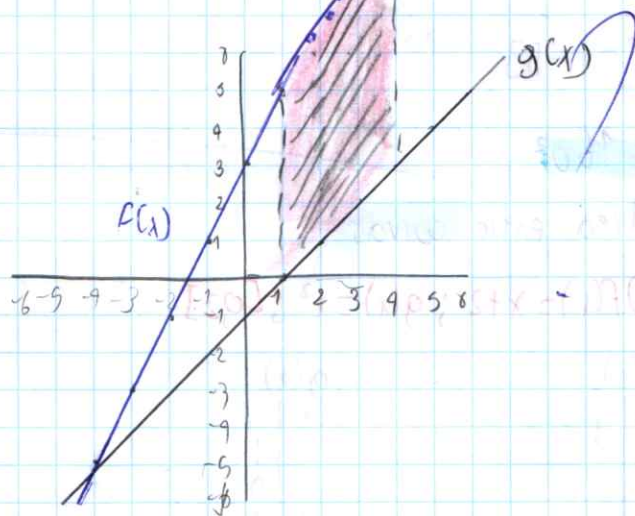
$$\int_{-\sqrt{2}}^{\sqrt{2}} (4 - 2x^2) dx = \frac{16\sqrt{2}}{3} \sqrt{2}$$

$$= 2 \int_0^{\sqrt{2}} (4 - 2x^2) dx$$

③ $f(x) = 2x + 3$; $g(x) = x - 1$; $[1, 4]$

x	y
1	5
2	7
3	9
4	11

x	y
1	0
2	1
3	2
4	3



$$(2x + 3) - (x - 1) = x + 4$$

$$A = \int_1^4 (x + 4) dx$$

$$= \left[\frac{x^2}{2} + 4x \right]_1^4$$

$$= \left(\frac{16}{2} + 16 \right) - \left(\frac{1}{2} + 4 \right)$$

$$= 24 - \frac{9}{2}$$

$$= \frac{48 - 9}{2}$$

$$= \frac{39}{2} \sqrt{2}$$

$$④ \int_0^2 (2x - x^2) dx$$

$$= \left[x^2 - \frac{x^3}{3} \right]_0^2$$

$$= 4 - \frac{8}{3}$$

$$= \frac{12-8}{3}$$

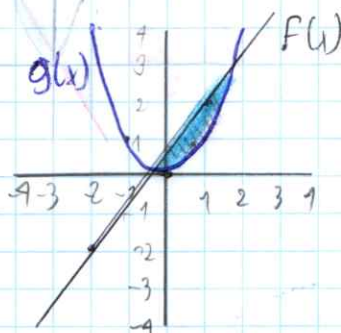
$$A = \frac{4}{3}$$

$$F(x)$$

x	y
-1	-2
0	0
1	2

$$g(x)$$

x	y
-1	1
0	0
1	1



$$⑤ \int_{-1}^0 (-x - x^2) dx + \int_0^1 (x - x^2) dx$$

$$= 2 \int_0^1 (x - x^2) dx$$

$$= 2 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1$$

$$= 2 \left(\frac{1}{2} - \frac{1}{3} \right)$$

$$= 2 \left(\frac{1}{6} \right)$$

$$A = \frac{1}{3} u^2$$

$$F(x)$$

x	y
-1	1
0	0
1	-1

$$g(x)$$

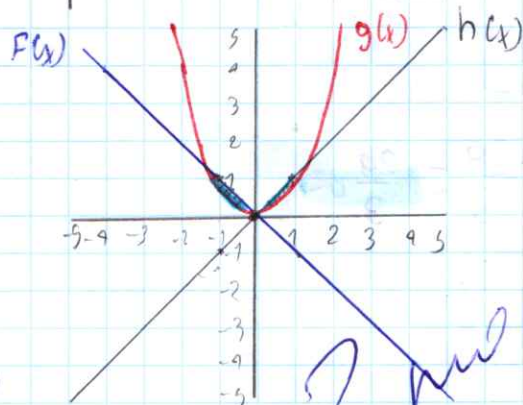
x	y
-1	1
0	0
1	1

$$h(x)$$

x	y
-1	-1
0	0
1	1

$$i(x)$$

x	y
-1	1
0	0
1	1



$$⑥ \sin x = \cos x$$

$$\tan x = 1$$

$$x = \frac{\pi}{4}$$

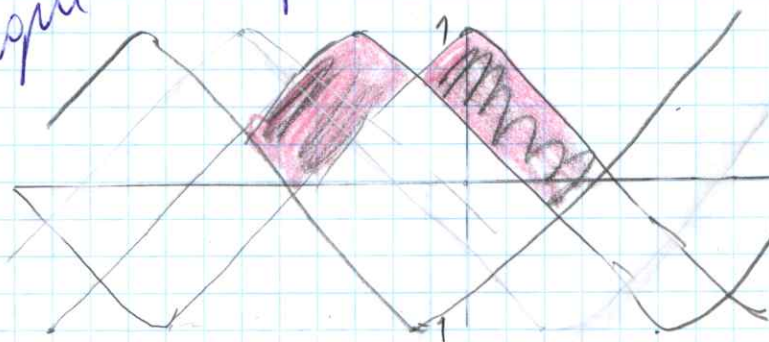
$$= [0, \frac{\pi}{2}]$$

$$\int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx$$

$$= [\sin x + \cos x]_0^{\pi/4}$$

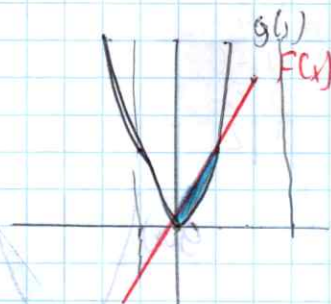
$$- [\cos x - \sin x]_{\pi/4}^{\pi/2}$$

$$A = 2(\sqrt{2} - 1)$$



① $f(x) = 2x$ y $g(x) = 2x^2$ $[1, 3]$

$$\begin{array}{c|c} f(x) & \\ \hline x & y \\ \hline 0 & 0 \\ 1 & 2 \\ 2 & 4 \\ 3 & 6 \\ -1 & -2 \end{array}$$

$$\begin{array}{c|c} g(x) & \\ \hline x & y \\ \hline -1 & 2 \\ 0 & 0 \\ 1 & 2 \\ 2 & 8 \end{array}$$


$$\int_1^3 [g(x) - f(x)] dx$$

$$\int_1^3 (2x^2 - 2x) dx$$

$$\int_1^3 2x^2 dx = \int_1^3 2x dx$$

$$\frac{2x^3}{3} \Big|_1^3 - \frac{2x^2}{2} \Big|_1^3$$

$$\left[\frac{2(3)^3}{3} - \frac{2(1)^3}{3} \right] - \left[\frac{2(3)^2}{2} - \frac{2(1)^2}{2} \right]$$

$$\left[\frac{54}{3} - \frac{2}{3} \right] - \left[\frac{18}{2} - \frac{2}{2} \right]$$

$$\frac{52}{3} - \frac{16}{2} = \frac{52}{3} - \frac{28}{3} = \frac{24}{3} = 8$$